CYS 331 —Unit 1: Wireless Technology

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Learning Outcomes

- Define wireless technology and explain why organizations value mobility, flexibility, and rapid connectivity.
- Differentiate wireless data networks, personal area networks, and wireless local area networks.
- Compare major transmission media and discuss their practical strengths, limitations, and risks.
- Describe wireless local area network security controls, countermeasures, and Wired Equivalent Privacy.

Introduction to Wireless Technologies

- Wireless technology transmits voice, data, or video without continuous physical cable connections.
- It uses electromagnetic waves through air, space, or materials to carry signals.
- Common wireless systems include local wireless networks, cellular services, Bluetooth, satellites, and tags.
- The main benefits are mobility, quick deployment, scalability, and support for portable devices.

Wireless Data Networks

- Wireless data networks connect devices using shared radio channels instead of network cables.
- They may cover rooms, buildings, cities, or countries depending on design.
- Performance depends on bandwidth, latency, interference, signal strength, and user density.
- Examples include campus hotspots, cellular internet, sensor networks, and fixed wireless broadband.

Personal Area Networks

- Personal area networks connect nearby personal devices within a very short range.
- They commonly support headphones, keyboards, smartwatches, health trackers, and file transfers.
- Bluetooth is most common because it balances convenience, energy efficiency, and affordability.
- Security still matters because trusted pairings can be abused or impersonated.

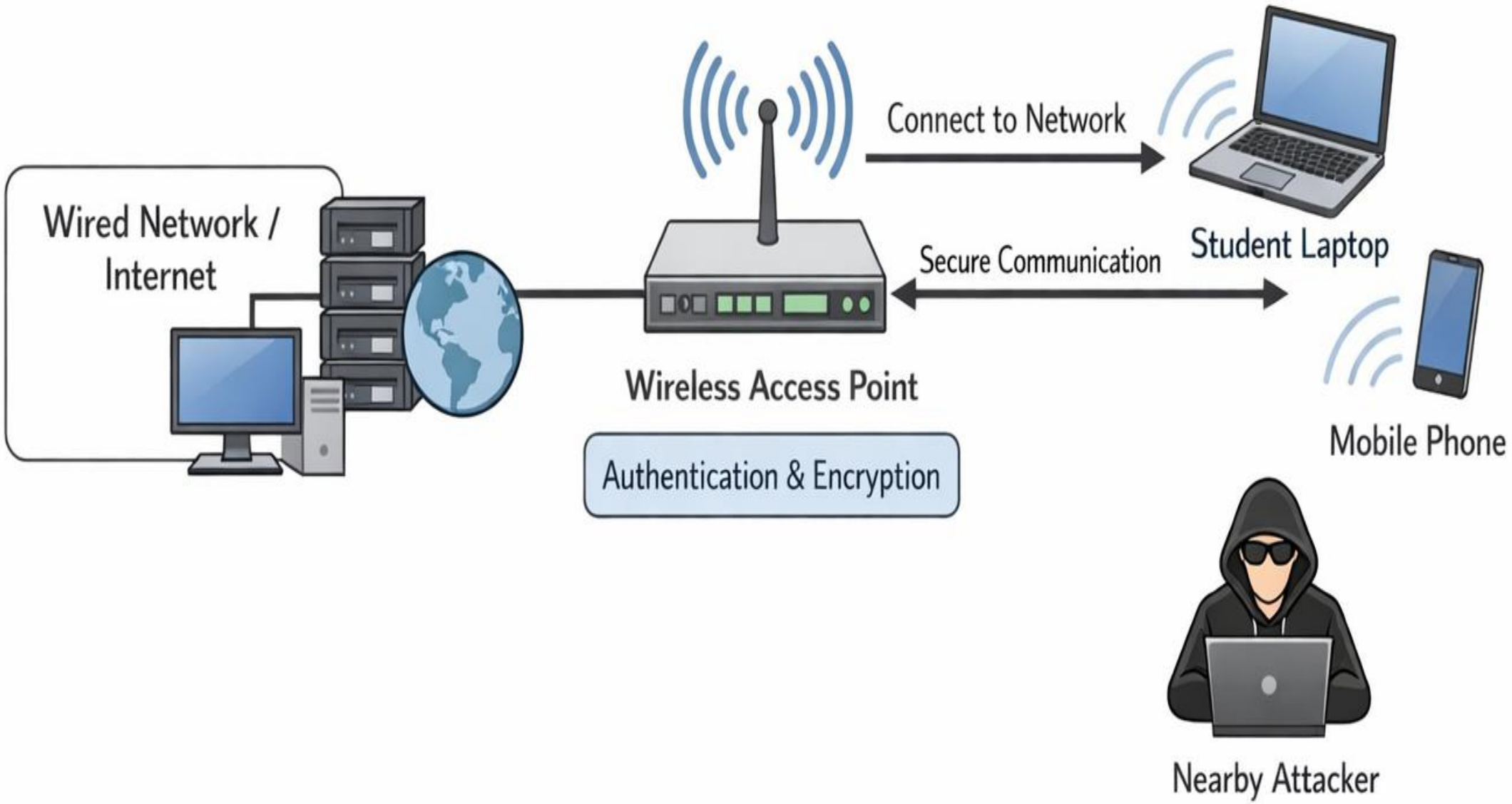
Transmission Media for Wireless Communication

Medium	Best feature	Main limitation	Typical example
Radio waves	Good coverage through rooms and buildings	Shared spectrum can face interference and eavesdropping	Wireless local area networks and broadcast services
Microwaves	High-capacity point-to-point links over distance	Usually need clearer line of sight	Backhaul between towers or campuses
Infrared	Useful for short, contained, low-interference links	Very short range and blocked easily	Remote controls and some room devices
Satellite	Extremely wide geographic reach	Higher latency, cost, and weather sensitivity	Remote internet and navigation services

Wireless Local Area Network Standards Overview

Standard	Band focus	Main contribution	Teaching note
802.11b	2.4 gigahertz	Made early wireless local area networking widely practical	Legacy technology with low speed and high interference exposure
802.11g	2.4 gigahertz	Improved speed while keeping broad compatibility	Still shares crowded spectrum with many household devices
802.11n	2.4 and 5 gigahertz	Used multiple antennas for better throughput and range	Important transition toward modern enterprise deployments
802.11ac	5 gigahertz	Increased capacity with wider channels and faster data rates	Strong fit for high-speed indoor networks
802.11ax	2.4 and 5 gigahertz, plus 6 in extended deployments	Improved efficiency for dense environments with many users	Useful in campuses, stadiums, and busy offices

How a Wireless Local Area Network Works



Securing Wireless Local Area Networks

- Strong passwords, modern encryption, and secure configuration are the foundation of protection.
- Change default settings, disable unused features, and separate guest access from internal systems.
- Limit signal exposure, update firmware regularly, and monitor logs for anomalies.
- Use layered controls because no single safeguard prevents every wireless attack.

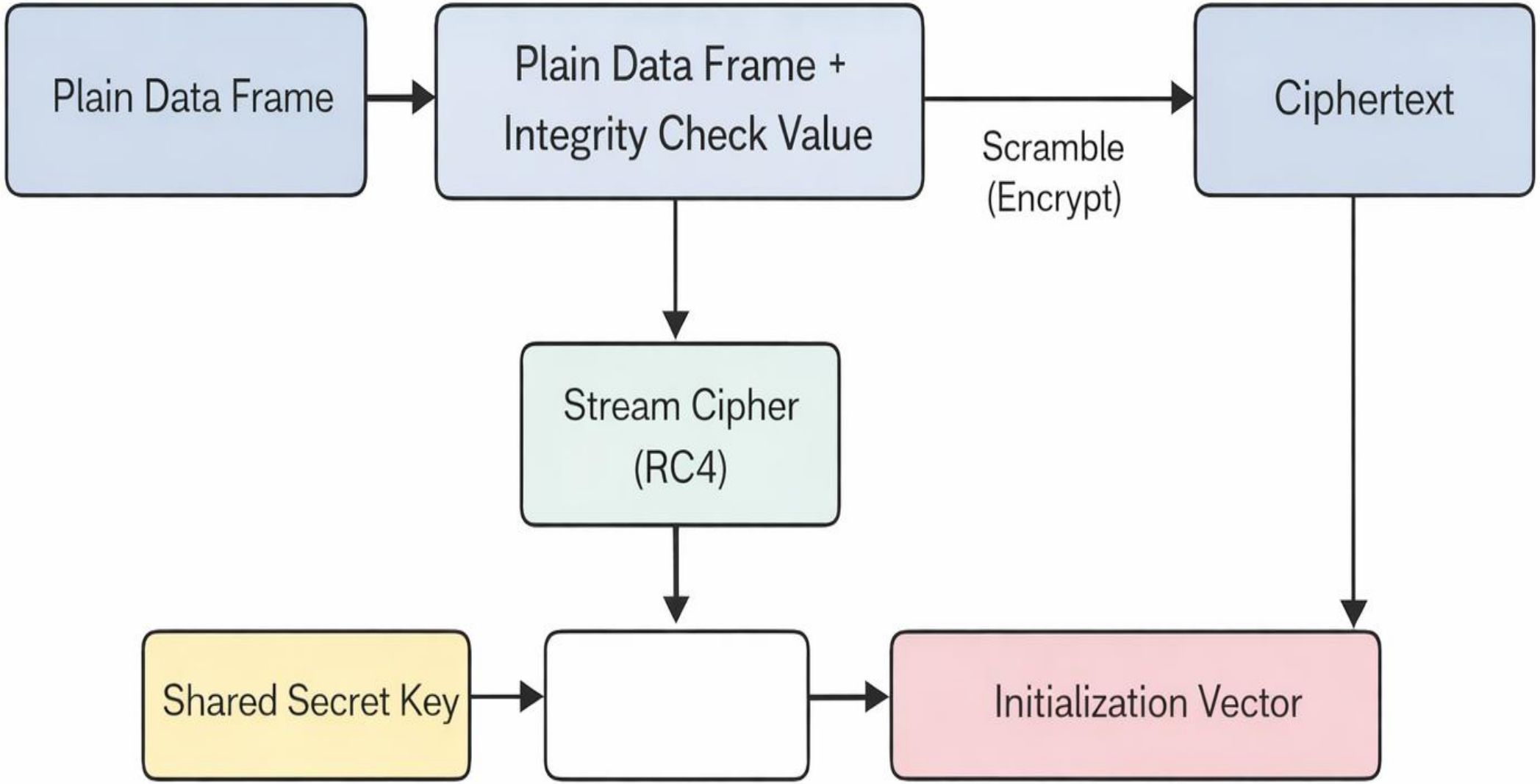
Countermeasures Against Wireless Attacks

- Site surveys help identify rogue devices, weak coverage, interference, and unsafe placement.
- Intrusion detection tools reveal suspicious associations, spoofing, deauthentication, and unusual traffic.
- Network segmentation limits damage when attackers gain access through wireless endpoints.
- User awareness reduces risky behavior such as sharing passwords or trusting unknown hotspots.

Wired Equivalent Privacy Fundamentals

- Wired Equivalent Privacy was an early wireless security mechanism for confidentiality.
- It combines a shared secret key, initialization vector, and stream cipher.
- WEP also adds an integrity check value to detect accidental changes.
- Designers wanted security comparable to wired networks, but implementation weaknesses remained.

Simplified Wired Equivalent Privacy Protection Flow



Why Wired Equivalent Privacy Is Weak

- Short initialization vectors repeat quickly, allowing attackers to analyze captured traffic.
- Static shared keys make compromise widespread because many devices reuse one credential.
- Weak integrity protection enables bit-flipping attacks and does not stop tampering.
- Modern networks should use wireless protected access standards instead of Wired Equivalent Privacy.

Unit 1 Summary

- Wireless technology improves mobility, but convenience must always be balanced with security.
- Understanding network types and media helps administrators choose suitable deployments.
- Wireless local area network standards shape compatibility, speed, and protective options.
- Secure design, countermeasures, and better protocols replace weak historical protections.